Flow Through Sudden Expansion in a Microchannel: A Numerical Study.

Fundamentals of Microscale Flows; ME 224

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Abstract

A detailed numerical investigation of steady incompressible laminar flow through a circular microchannel with sudden expansion is presented. The study focuses on analysing the influence of Reynolds number and expansion ratio on flow separation characteristics, reattachment length, friction factor and pressure loss behaviour. A two-dimensional axisymmetric computational model was developed in ANSYS Fluent based on the finite volume method. Simulations were performed for Reynolds numbers ranging from 1 to 100 for expansion ratios of 1.51 and 2. The incompressible Navier–Stokes equations were solved under steady-state conditions with no-slip wall boundaries. Results demonstrate nonlinear growth of reattachment length with Reynolds number due to strengthening inertial effects and adverse pressure gradients. The friction factor decreases inversely with Reynolds number, consistent with classical laminar theory. Numerical trends are in strong agreement with previously reported experimental and computational studies on microchannel sudden expansion flows, confirming applicability of continuum hydrodynamics at the microscale within the investigated regime.

Content

1. Introduction

Microfluidic systems operate in channels with characteristic dimensions in the micrometer range, where viscous forces dominate, and flow remains predominantly laminar. Despite the small scale, geometric discontinuities such as sudden expansion significantly alter the flow field by introducing strong adverse pressure gradients.

When fluid flows through a sudden expansion:

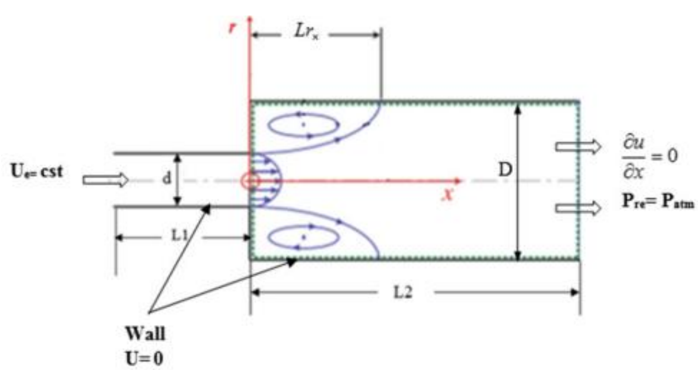
• The sudden increase in cro ss-sectional area causes a rapid pressure rise.
• The pressure increase generates an adverse pressure gradient.
• The boundary layer detaches from the wall.
• A recirculation zone forms downstream of the expansion.

The separated shear layer eventually reattaches to the wall at a downstream location defined as the **reattachment length (Lr)**.

Understanding these flow characteristics is essential because:

• They contribute to additional head losses.
• They influence residence time and mixing.
• They affect microfluidic device efficiency.

The present study numerically investigates the hydrodynamic behaviour of laminar flow through sudden expansion in microchannels, focusing on the effect of Reynolds number and expansion ratio.



2. Literature Review

Extensive investigations have been conducted to understand expansion flows in both macro and microchannels.

Djellouli and Filali (2022), numerically analyzed water flow through circular microchannels with expansion ratios 1.51, 2 and 3 for Reynolds numbers up to 1000. Their findings revealed:

• Reattachment length increases exponentially at low Reynolds numbers and linearly beyond a critical Reynolds number.

• Friction factor follows $\lambda \approx 64/Re$ in laminar regime.

• Singular pressure drop coefficient decreases sharply at low Re and stabilizes beyond a critical value.

Experimental studies by Khodaparast et al. confirmed that microchannel expansion flow exhibits similar separation behaviour as conventional-scale laminar expansion flow, validating continuum mechanics assumptions at microscale.

These studies establish a strong theoretical and experimental foundation for the present investigation.

3. Motivation and Objective

Although classical hydrodynamics predicts laminar expansion behavior, microscale flows may exhibit deviations due to enhanced viscous dominance and surface interactions.

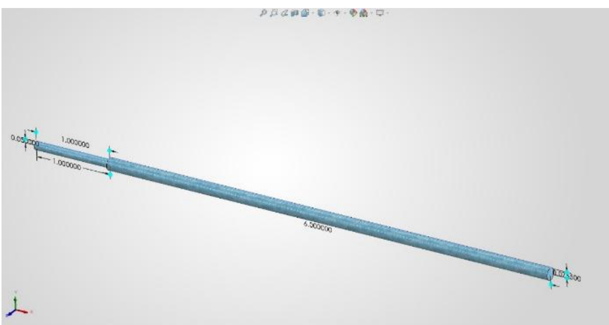
The motivation of this work is to examine whether classical flow trends remain valid for low Reynolds number microchannel flows with moderate expansion ratios.

Objectives:

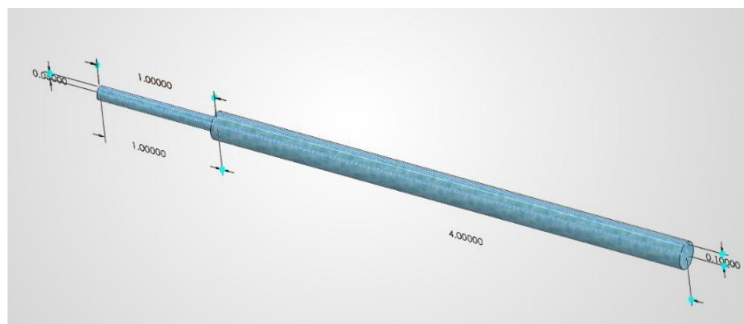
- To numerically simulate steady laminar flow through sudden expansion microchannel
- To quantify reattachment length variation with Reynolds number
- To determine friction factor behaviour
- To examine influence of expansion ratio
- To compare numerical trends with published literature

Reynolds numbers considered: 1, 20, 50, 70, 100

Expansion ratios: Rp = 1.51 and Rp = 2



Expansion ratio:1.51



Expansion ratio:2

4. Methodology

4.1 Governing Equations:

The incompressible steady-state Navier–Stokes equations govern the flow:

Continuity:

$$\frac{\partial \left(u_i\right)}{\partial x_i} = 0$$

Momentum:

$$\frac{\partial\left(u_i u_j\right)}{\partial x_i} = -\frac{1}{\rho_{\mathrm{nf}}}\frac{\partial P}{\partial x_j} + \nu_{\mathrm{nf}}\frac{\partial}{\partial x_i}\left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right)$$

Reynolds number:

$$\mathrm{Re} = \frac{\rho U_m D_h}{\mu}$$

Reattachment Length:

$$L_r = \frac{L_{rx}}{d}$$

Friction Factor:

$$\lambda = \frac{\Delta P_f D}{\frac{1}{2}\rho U^2 L}$$

4.2 Numerical Method:

The finite volume method was used to discretize governing equations. Pressure–velocity coupling was handled using the SIMPLE algorithm. Second-order spatial discretization was employed for improved accuracy.

Convergence was ensured when residuals dropped below 10^{-5} for all governing equations.

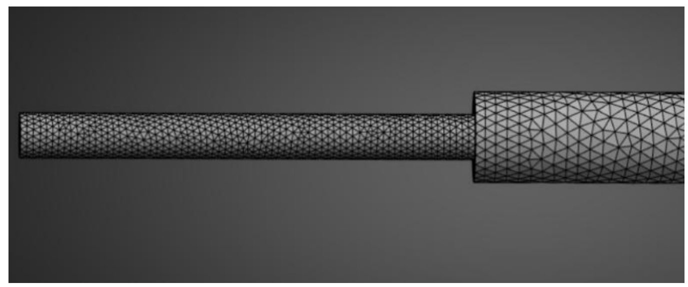
4.3 Mesh Independence Study:

To ensure numerical accuracy, mesh refinement was performed near the expansion plane and wall regions where strong gradients occur.

Three progressively refined meshes were tested. Variation in reattachment length between medium and fine mesh was below 2%, confirming mesh independence of results.

This ensures that reported trends are physically meaningful rather than numerical artifacts.

Expansion Ratio:1.51



Expansion Ratio:2

4.4 Boundary Conditions:

- Velocity inlet corresponding to desired Reynolds numbers
- Pressure outlet at atmospheric pressure
- No-slip condition at walls

Working fluid: Water at 293.15 K

$\rho = 998.2\ kg/m^3$

$\mu = 0.001003\ kg/m \cdot s$

5. Results and Discussion

5.1 Flow Structure and Separation:

Velocity streamline contours reveal development of symmetric recirculation zones immediately downstream of the expansion.

At Re = 1:

• Viscous forces dominate

• Flow remains largely attached

• Recirculation zone is negligible

As Reynolds number increases:

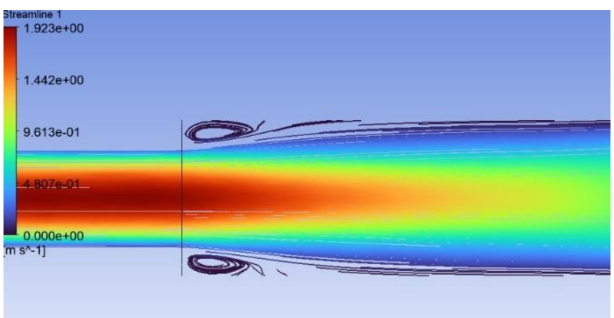
• Inertial effects intensify

• Adverse pressure gradient strengthens

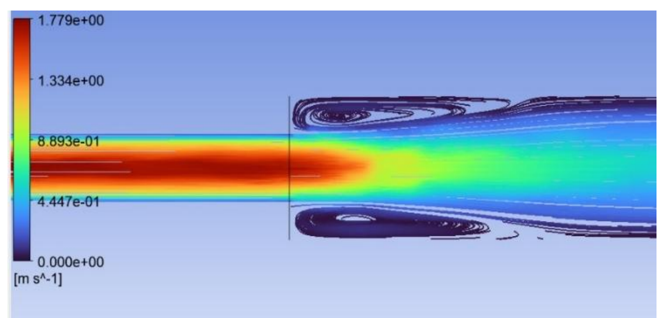
• Separation becomes more pronounced

• Recirculation region enlarges

The growth of separation zone is attributed to increasing dominance of inertial forces over viscous diffusion.



Streamlines at Re = 100 for Rp=1.51



Streamlines at Re=100 for Rp=2

5.2 Reattachment Length Behavior:

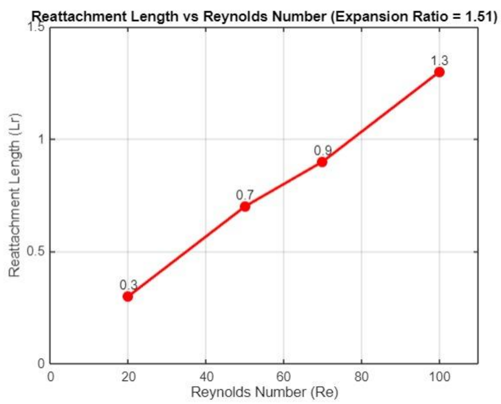
Reattachment length increases nonlinearly with Reynolds number.

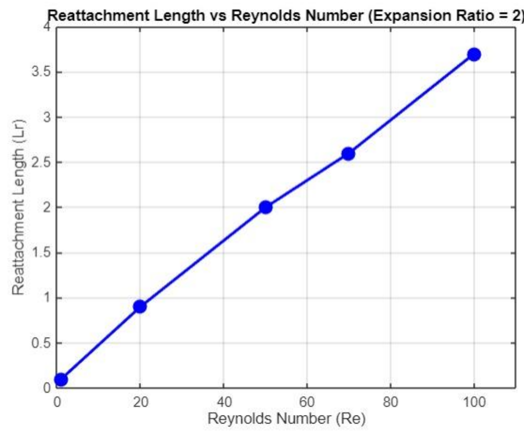
For low Reynolds numbers (Re < ~20):

• Growth rate is gradual due to strong viscous damping.

Beyond this range:
• Increase becomes more rapid due to stronger inertial contribution.

Higher expansion ratio (Rp = 2) produces larger reattachment length due to stronger pressure recovery and adverse pressure gradient.



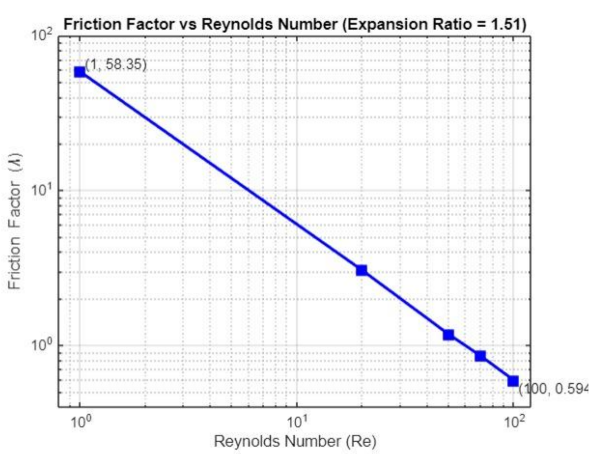


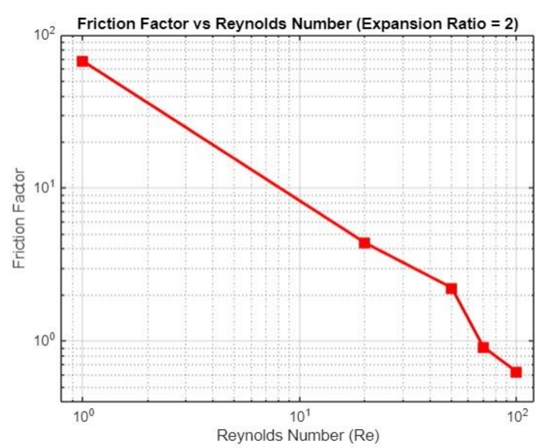
5.3 Friction Factor Behaviour:

Friction factor decreases inversely with Reynolds number.

This trend can be explained by relative force dominance:
Low Re → viscous forces dominate → high wall shear stress → larger λ.
Higher Re → inertial forces dominate → reduced relative viscous resistance → lower λ.

Numerical values closely follow theoretical laminar relation λ = 64/Re, confirming negligible scale-induced deviation within studied regime.





5.4 Comparison with Literature:

The nonlinear increase in reattachment length and inverse dependence of friction factor on Reynolds number are consistent with numerical and experimental observations reported in literature.

The close agreement confirms validity of continuum-based Navier–Stokes equations for laminar microscale expansion flows.

6. Conclusions

A comprehensive numerical investigation of laminar flow through sudden expansion microchannel was conducted.

Key conclusions:

- Reattachment length increases nonlinearly with Reynolds number.
- Larger expansion ratio results in stronger separation and longer recirculation zone.
- Friction factor decreases inversely with Reynolds number.
- Numerical results agree with classical laminar hydrodynamic theory.
- Continuum modelling remains valid for studied microscale laminar.

7. Further Scope

Future research may include:

- Extension to higher Reynolds numbers approaching transition.
- Three-dimensional modelling.
- Inclusion of heat transfer effects.
- Investigation of non-Newtonian biofluids.
- Experimental validation.

References

1. Djellouli, S. and Filali, E.G., 2022. Numerical analysis of the dynamic aspects of flow through a microchannel with sudden expansion. Desalination and Water Treatment, 279, pp.78–84.
2. Zhang, J. et al., 2022. Experimental and Numerical Studies of Liquid-Liquid Two-Phase Flows in Microchannel with Sudden Expansion/Contraction Cavities. Chemical Engineering Journal.